

I design and implement complex software systems: modular .NET platforms, backend services with payments and recovery, compilers, and program-analysis tools. My strongest areas are explicit contracts, state invariants, verifiable semantics, and carrying systems through to reproducible outcomes.

Platform architecture

UniversalToolchain/Wist2: independent packages, typed contracts, routes, manifests/locks, and lifecycle ownership

Production backend

Payments, subscriptions, roles, devices, PostgreSQL/SQLite, migrations, recovery, backup/restore, and rollback

Compiler / systems

Callable-first SSA, interpreter/CIL, LLVM, C++23, memory analysis, and x86-64 code generation

EXPERIENCE		SKILLS
2024 — present	UniversalToolchain / Wist2 — creator and primary developer - Designed language composition from independently versioned packages: typed artifacts, capabilities and conflicts, deterministic routes, and pass ordering. - Implemented exact manifest binding, lock identity, fail-closed policies, and runtime-session lifecycle ownership.	Architecture module boundaries, typed contracts, state machines, manifests/locks, lifecycle ownership, fail-closed design .NET / backend C#/.NET, ASP.NET Core, REST/OpenAPI, PostgreSQL, SQLite, webhooks, transactions, migrations, recovery Compilers Bytecode/AIR, CFG, SSA, dominance, optimizations, interpreter/CIL, LLVM, program analysis Systems and verification C++23, C17, Rust, Python, Linux, CMake, sanitizers, differential/metamorphic testing, exact oracles
July 1 — August 31, 2026	MCST — compiler engineering intern Developing an LLVM optimization pass and implementing tested graph algorithms in C++23.	
2023 — 2025	Mama Lama — Software Developer - Independently developed games and applications for DJs and events, from requirements and design through implementation, testing, and preparation for use. - Designed the application architecture and modular structure, defining component boundaries and interfaces; owned the complete technical implementation through a finished product.	
PROJECTS		EDUCATION
UniversalToolchain / Wist2 1,465/1,465 tests with zero failures across nine packages and clean-consumer projects, plus interpreter/CIL parity and reproducible configuration checks. VpnMediator A production service with controlled state transitions, backup/restore, health gates, and safe rollback; the public case study excludes secrets and user data. PS-form Memory Dependence Analyzer Ranked 1st of 49 with 104/104 tests, with reproducible wrong-answer and time-limit counterexamples.		Software Engineering student at HSE University
		RECOGNITION
		HSE Vysshaya Proba prize-winner in competitive and industrial programming; regional Russian informatics olympiad prize-winner. Moscow / remote